

Machine learning engineer researching and building medical AI products for 4 years. Experienced in full-cycle model development for chest X-ray and lung CT, product deployment and optimization, and designing reusable data and evaluation pipelines.

Experience

ML Engineer

VUNO

Mar 2022 – Present (4 yrs 7 mos)

Chest X-ray Team

Sep 2024 – Present

- Diagnostic AI products for chest X-ray (VUNO Med®-Chest X-ray™)
- Model R&D, ONNX/OpenVINO serialization, and deployment optimization
 - Built reusable DB-based training and evaluation pipelines with standardized metrics
 - Developed internal developer tools and established code review practices

Lung CT Team

Nov 2022 – Aug 2024

- Diagnostic AI products for lung CT (VUNO Med®-LungCT AI™)
- Built shared 3D lung imaging pipelines to eliminate duplicate code across models
 - Quantitatively analyzed label errors in external evaluation datasets to guide relabeling by overseas hospitals
 - Introduced team MongoDB for data integrity and performed on-site hospital validation

Digital Oncology Team

Mar 2022 – Oct 2022

- Digital pathology research and development
- 1st place in TIGER Challenge survival analysis track (inference pipeline & TIL analysis) ^[1]
 - Built containerized pipeline connecting multi-stage models and released open-source
 - Designed modular reusable library for whole slide image (WSI) processing

Researcher

HVCL @ Korea Univ.

Jun 2019 – Feb 2022

Interactive Segmentation Research

- Deep learning models and interactive tools to reduce physician annotation effort ^[C1]

Organized International Medical Data Challenges (2 Editions)

- Web tool design, implementation, and qualitative evaluation for 3 hospitals ^[U2]
- Built and operated a digital platform for sharing large-scale pathology datasets ^[U3]

Skills	ML & Serving	Python, PyTorch, ONNX, OpenVINO
	Data & MLOps	MongoDB, Airflow, MLflow, W&B
	DevOps & CI/CD	Docker, Git, GitHub Actions
	AI Tools & Agents	Claude Code, Codex, Gemini

Education	M.E. in CS	Korea University	2020 – 2022
	B.S. in CS	UNIST	2015 – 2020
Awards	1st Place, International Pathology Image Challenge (TIGER Challenge) Radboud University		2022
	1st Place, Korea University Startup Competition Seoul Metropolitan Government, Seongbuk-gu, SH Corporation, Korea University		2021
	1st Place, Dental Panoramic X-ray AI Challenge Healthhub Datathon Track B, Seoul National University Dental Hospital		2020
Publications	JOURNALS		
	[J1]	Analysis of computational tumor-infiltrating lymphocytes in breast cancer from the results of the TIGER challenge TIGER Challenge Consortium Nature Communications	2025
	[J2]	PAIP 2020: Microsatellite instability prediction in colorectal cancer Kyungmo Kim*, Kyoungbun Lee*, <u>Sungduk Cho*</u> , Dong Un Kang, Seongkeun Park, et al. Medical Image Analysis	2023
	[J3]	Development and operation of a digital platform for sharing pathology image data Yunsook Kang, Yoo Jung Kim, Seongkeun Park, Gun Ro, Choyeon Hong, Hyungjoon Jang, <u>Sungduk Cho</u> , et al. BMC Medical Informatics and Decision Making	2021
	CONFERENCES		
	[C1]	DeepScribble: Interactive Pathology Image Segmentation Using Deep Neural Networks with Scribbles <u>Sungduk Cho</u> , Hyungjoon Jang, Jing Wei Tan, Won-Ki Jeong IEEE International Symposium on Biomedical Imaging (ISBI)	2021
	PREPRINT		
	[P1]	Joint-Embedding Predictive Architecture for Self-Supervised Learning of Mask Classification Architecture Dong-Hee Kim, <u>Sungduk Cho</u> , Hyeonwoo Cho, Chanmin Park, Jinyoung Kim, Won Hwa Kim arXiv	2024
	* Equal contribution		
Languages	English	OPIC Advanced Low	Jul 2025